Journal of Open Innovation: Technology, Market, and Complexity 11 (2025) 100508

Contents lists available at ScienceDirect

Journal of Open Innovation: Technology, Market, and Complexity

journal homepage: www.sciencedirect.com/journal/journal-of-open-innovation-technology-market-and-complexity

Analyzing AI regulation through literature and current trends

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ARTICLE INFO

Keywords:
AI regulation
AI Regulatory frameworks
Artificial Intelligence policies
Future trends AI regulation
Comparative study AI regulation
AI Governance mechanisms
EU AI Act

ABSTRACT

Artificial intelligence (AI) technology has been rapidly evolving, necessitating that discussions on the need and design of regulatory frameworks be taken seriously. This paper reviews literature regarding AI regulations on a theoretical versus a practical level. It examines the different models of regulation – some of which are risk-based regulation and others are complete prohibition – to gauge the literature's predictions on the scope, precise features, and direction of AI regulation. With particular reference to the EU AI Act as the primary case study, the paper analyzes the consequences of rules on innovation and the international standards' impact on AI regulatory measures. The results show that almost all scholarly assumptions are accurate, showing how the issues are integrated at the practical level and where is it still challenging to comply and enforce. This paper advocates for continuous improvement in AI regulatory frameworks and more international interactions to ensure efficient governance for AI technologies. It contributes to ongoing debates about creating future-proof adaptable robust AI regulations which can negotiate the complexity between technological development and social protection.

1. Introduction

The implementation of Artificial Intelligence (AI) across the globe has resulted in intense debate about the need and norms of regulative frameworks that manage its impacts. It is important to know how well the existing literature captures the future trends, scope, and characteristics of AI regulation to ensure that academic insights are aligned with real and anticipated regulatory provisions. In this regard, our study is a systematic review of how theoretical discourses on AI have matched actual implementation considering issues at stake, their aims and their governing structures using preexisting and novel methods.

In relation to AI, the regulatory landscape is in a state of constant change and domestically implemented by different governing bodies. These include frameworks based on risk, specific bans and classification systems created to address the unique challenges arising from AI technologies. These themes have been widely studied in the literature, and they help provide a basis for gauging how much of these academic speculations are reflected in practical regulatory practices. For example, Chauhan (2022) examined AI regulation as an imperative-driven approach, discussing the procedural mechanics of regulatory proposals along with their practical implications (Chauhan, 2022). Hacker (2020) examined the risk-based approaches to regulation highlighting the importance of legal flexibility and certainty in governing AI in the European context (Hacker, 2020).

This review also examines the anticipated and actual impacts of AI regulation on innovation, international regulatory frameworks, and the accompanying dangers and controversies. According to Dumouchel (2023), flawed conceptions regarding the definition of AI can lead to

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https://doi.org/10.1016/j.joitmc.2025.100508
Received 7 October 2024; Received in revised form 22 February 2025; Accepted 26 February 2025
Available online 28 February 2025
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inadequate regulatory practices while contextualizing appropriate regulatory approaches (Dumouchel, 2023). Schuett, however, underlines the need for an all-encompassing regulatory approach and manages to adequately address the recurring issue of the balance between regulation and innovation in the literature on the AI Act– risk management provisions (Schuett, 2023).

This study seeks to bridge the divide between academic foresight and regulatory practices by analyzing existing literature since 2020. It examines the extent to which purported theoretical systems correlate with actual or proposed actions, within global AI governance, thereby evaluating the effectiveness of existing regulatory frameworks in solving AI's multi-faceted challenges. Consequently, it contributes to the debate on the need to establish politically and socially acceptable regulatory frameworks for AI which, together with the advancement of innovation, ensure societal safety.

1.1. Objective of this study

This research will assess the extent to which the existing literature on AI regulation and its current regulatory measures are aligned by looking at how AI regulations augment in real life, their predicted orientations, scope and other characteristics. It also considers whether the previous conversations about AI system classification and restrictions match risk-based approaches and specific regulation prohibitions. Moreover, it reviews the possible impacts of AI regulations on innovation and global standards while enumerating the challenges and criticisms faced so far.

The study aims to find and synthesize literature that answers the following research questions.

1. How accurately do the current literature anticipate the direction, scope, and features of AI regulation, including concerns, objectives, and governance mechanisms, compared to actual and forthcoming regulatory provisions?
2. In what ways do the risk-based approach and specific prohibitions in current AI regulations align with previous discussions on AI system classification and anticipated restrictions?
3. How do the predicted and actual impacts of AI regulation on innovation, global regulatory standards, and the anticipated challenges and criticisms of AI regulation compare to prior literature?

Problem Statement:

The growing field of AI moderation poses a crucial requirement for assessing the congruence between theoretical frameworks and scholarly forecasts with actual regulatory actions, including those that are yet to be enforced and the plausible consequences that they hold for innovation and worldwide harmonization of regulation.

2. Materials and methods

2.1. Criteria for selecting the studies

To gain insights into the existing governing structures in the management of AI, a systematic approach was adopted in this study.

Inclusion criteria:

- Empirical scholarly articles published between 2020 and 2024 that have undergone peer review.
- Articles that concern frameworks of governance, the general abilities of AI regulation, or the impact of AI regulation on innovation.
- Articles addressing certain regulatory approaches, particularly those addressing risk-adjusted regulation or regulatory compliance.

Exclusion criteria:

- Non-peer-reviewed articles, commentaries, and other publications that did not specifically focus on AI regulation.
- Articles that were published in any other languages apart from English.
- General studies with a broader scope of technology do not focus on AI regulations.

2.2. Data sources and search strategy

This review considered sources such as Scopus and Google Scholar. When placing the search, Boolean operators were utilized to narrow down the results, for example "AI Regulation" AND "Governance Frameworks" or "Risk Based AI Governance" AND "Innovation." It was ensured that the topic was covered entirely without losing the focus of the relevant studies.

2.3. Data analysis

The thematic bifurcation of articles provided a framework for examining the relevant literature. The recurring themes identified included risk based regulation, ethical concerns, and innovation impacts. This helped in extracting details, including cross-cutting regulatory frameworks (national, regional, and global), governance systems (compliance, classification, and other measures), and consequences and changes in innovation and social issues.

2.4. Review process

The comprehensive review process is described visually in Figure 1, and illustrates the methods used to include studies, detailing the steps starting from initial article selection until the studies that were finally considered in the review.

3. Review of literature / results

3.1. How accurately do the current literature anticipate the direction, scope, and features of AI regulation, including concerns, objectives, and governance mechanisms, compared to actual and forthcoming regulatory provisions?

3.1.1. Direction and scope of AI regulation

The literature on AI regulations is keenly concerned with the direction and extent that should be taken to accommodate changes in technology. Highlighting the fragmented state of AI governance research, Birkstedt et al. (2023) emphasized the need for a coherent framework that considers technological, stakeholder and contextual dimensions. This approach shows that regulatory measures ought to be aligned with the complicated and multifaceted nature of AI applications (Birkstedt et al., 2023). Similarly, Chauhan (2022) proposed an imperative-driven framework that clarifies the defining parameters of AI for regulatory purposes to ensure a well-defined targeted regulation scope (Chauhan, 2022).

3.2. Critical analysis: Alignment between theoretical predictions and EU AI Act

The EU AI Act aligns with the scholarly expectations on risk classification and ethical management on AI use, but not for generative AI. Academic models seek to emphasize dynamic AI systems, providing the examples of real-time generation and modification of content, which the Act does not provide for. This counterpoint reveals a worrying missing link, particularly especially when newer technologies such as ChatGPT require completely different sets of regulations. On the one hand, the Act's focus on high-risk systems deals with ethical issues that AI poses, but the regulatory framework is too rigid for fast-moving AI systems; hence it needs to be constantly updated (Schuett, 2023; Stepanyan, 2021)

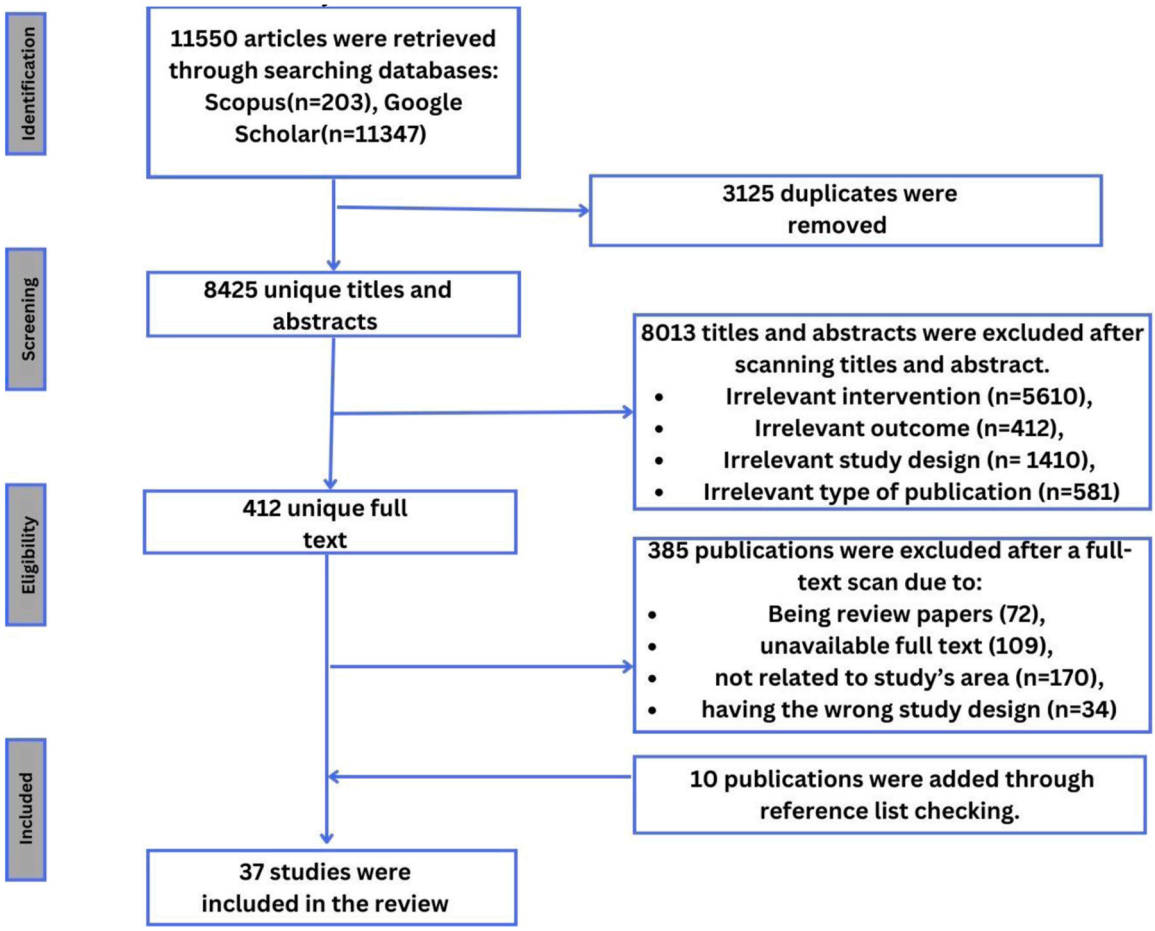

Fig. 1. Review methodology.

3.2.1. Concerns and objectives of AI regulation

The central concerns driving AI regulation have based on ethical, safety and societal implications. Taeihagh (2021) emphasizes that the advantages and risks AI brings are unanticipated hence, it requires strong governance. Additionally, these issues are of immense importance when considering the intent behind AI regulation which is to maximize calls and minimize threats (Taeihagh, 2021). Morley et al. (2022) also pointed out collaboration across borders in matters of safety and ethics, which suggests the need to regulate AI using universal principles (Morley et al., 2022).

3.2.2. Governance mechanisms

AI governance is critical as it requires the combination of stakeholder engagement, institutional arrangements and regulatory frameworks. For instance, Almeida et al. (2023) suggest a special multi-level governance structure that integrates ethical considerations and stakeholder participation in the development and deployment of AI technologies (Almeida et al., 2023). Schneider et al. (2020) also support this suggestion and call for governance frameworks that integrate, "data and models within systems engineering for risk mitigation and increased accountability of AI applications" (Schneider et al., 2020).

3.2.3. Risk-based approaches

There is an overarching focus in recent literature concerning the regulation of AI monopolies that implement risk-based methods. Stepanyan (2021) analyzed the European AI Act, which serves as a regulatory framework for the categorization of risk control systems through the close supervision of high-risk AI systems (Stepanyan, 2021). This can also be seen in Schuett's (2023) work, in which their research explains the particular details of a risk management provision that makes safe application of Artificial Intelligence using Schuett's explanation (Schuett, 2023).

3.2.4. Anticipated restrictions

Certain risks associated with AI technologies are meant to be reduced while developing regulations around AI Mitigating specific risks related to AI technologies, as suggested by Smuha (2020), where the author elaborates on the ethical elements and policies designed to control bias, privacy, and accountability in AI systems (Smuha, 2020). This aligns with the overarching objectives of policy-making that seek to foster and safeguard the responsible functioning of AI systems with respect to society.

3.2.5. Impact on innovation

There is a complicated balance between AI regulation and innovation. Different regulatory strategies can assist or obstruct technological developments. Kurshan et al. (2020) analyze the challenges of AI model governance within the financial services industry and note that there must be a balanced policy structure that encourages innovations while simultaneously ensuring compliance and risk mitigation at the same time (Kurshan et al., 2020). In other words, Hacker (2020) also argues for an indefinite AI control that advances technology while considering ecological and other societal factors (Hacker, 2020).

3.2.6. Global regulatory standards

Global regulatory standards are fundamental for the harmonization of AI governance across different jurisdictions. Morley et al. (2022) suggested that there should be international standards and combinations

of efforts to effectively tackle the complex problems brought forth by AI technologies that are based on their assertions (Morley et al., 2022). Birkstedt et al. (2023) provided similar reasoning in support of broader contextual investigations and a robust understanding of AI governance frameworks to achieve rational global norms (Birkstedt et al., 2023).

3.2.7. Anticipated challenges and criticisms

It is important to consider implementation, efficiency, and impact on society while attempting to create regulatory frameworks for AI. Wirtz et al. (2020) mention in their paper the difficulties in public administration coping with the pace of AI progression and the need for comprehensive regulations to mitigate potential risk (Wirtz et al., 2020). Moreover, Stark et al. (2020) argued that AI governance mechanisms can be captured by corporate and state interests which require more active and equitable governance systems (Stark et al., 2020 ‘).

The latest scientific data explain the further characteristics and extent of AI regulation that has to be done. It also points out the need for ethical, risk-based, and global systems of governance that can manage the multitude of consequences caused by AI. While these approaches face challenges and dissenting opinions, they do make efforts towards mediating between the need for innovation and the social safety nets that enshrine responsibility for the development and deployment of AI technologies.

3.2.8. India’s perspective on AI regulation

Regarding the use of AI in India, NITI Aayog amends its policies to capture development and social provisioning. Unlike the risk-oriented model of the EU, India adopts an innovation-focused governance style that prioritizes employment as well as the responsible use of Artificial Intelligence,which is seen in the National Strategy on AI, where AI is intentionally provided for use in public sectors such as education and healthcare. India’s model sheds light on how regulatory frameworks are designed in emerging economies that balance growth with regulation, and simultaneously provides guidance on global AI governance (NITI Aayog, 2018).

3.3. In what ways does the risk-based approach and specific prohibitions in current AI regulations align with earlier discussions on AI system classification and anticipated restrictions?

Currently the cornerstone of AI regulation is risk-based approaches that aim to classify AI systems based on their risks and, consequently, put mechanisms of governance. This regulatory strategy is closely aligned with earlier academic discussions on the classification of AI systems and the resulting restrictions. The author further revealed that the European Union’s proposed Artificial Intelligence Act (AI Act) embodies a holistic approach in categorizing and managing AI systems based on their levels of risk (Schuett, 2023).

3.3.1. Historical context of AI system classification

Various models, in which AI systems have been historically classified, have emerged to address the ethical and operational challenges of these machines. According to Mökander and his colleagues, these models comprise three types: the binary classification model, the risk-based classification model called Ladder, and the multi-dimensional model commonly termed as matrix (Mökander et al., 2022). These models provide the basis for the current regulatory frameworks created with the goal of recognizing the underlying risks and ethical issues associated with various AI (Mökander et al., 2022).

3.3.2. Risk-based regulation and the AI Act

This classification resembles previous models regarding the AI Act’s risk-based regulatory approach. The AI Act enumerates some of the specific requirements for systems with high-risk AI, which include compliance and other forms of incrimination that are strict and covertly subscribe to accepted notions regarding the prevalence classification of AI risk theories (Hacker, 2020). This merging guarantees that regulatory efforts are directed towards managing the most significant dangers posed by AI technologies.

3.3.3. Specific prohibitions in AI regulations

Current rules surrounding AI allow for bans on certain types of AI activities aimed at safeguarding fundamental human rights and ensuring ethical conduct. Nesterova (2022) accuses the AI Act of not being sufficiently expansive to ban AI systems that may violate fundamental rights such as biometric surveillance technologies. This critique echoes earlier debates that called for explicit, enforceable limitations on high-risk AI applications to avert potential abuse (Nesterova, 2022).

3.3.4. Alignment with ethical and safety concerns

The key issues of the AI Act are ethical considerations and safety, which is in line with]previous academic attention to these matters. Risk-based approaches towards EU digital regulation seek to balance innovation and rights protection, according to De Gregorio and Dunn (2022), a theme often repeated in AI ethics and governance literature (De Gregorio and Dunn, 2022).

3.3.5. Practical implementation of risk-based governance

Practical measures, such as risk assessment and the effective development of compliance mechanisms are critical in the implementation of risk-based governance in AI. Regarding how the European Commission’s regulatory strategy is able to differentiate and manage AI systems through risk analysis, Paul (2023) states that high-risk systems must undergo thorough scrutiny (Paul, 2023). This is indicative of the necessity for a structured and systematic approach to operationalize AI governance.

3.3.6. Challenges and recommendations

Despite the concurrence with former talks, the risk-based approach implementation confronts different problems. Hauer et al. (2023) found that there are issues concerning clarity and applicability of regulatory provisions that require more refined formulations to ensure effective regulation (Hauer et al., 2023). These challenges signify the continuing requirement for refinement and readjustment in regulatory frameworks to cater to evolving threats.

In summary, the degree to which current AI regulations align with previous AI system classification talks and predicated limits demonstrates how risk-based models are essential for handling AI technologies. Current efforts to regulate highlight specific prohibitions and focus on ethics and safety concerns according to the theoretical underpinnings of earlier scholarly works. Nevertheless, there are still practical challenges that call for regular improvement and change of regulatory frameworks so that the governance of AI can accompany developments in technology.

3.4. How do the predicted and actual impacts of AI regulation on innovation, global regulatory standards, and the anticipated challenges and criticisms of AI regulation compare to prior literature?

There has been a great deal of debate over the effects of AI regulations on innovation, global regulatory standards possible challenges and criticism. This debate reflects a range of perspectives pointing to both proponents and opponents of AI regulation and its implications. By comparing forecasted impacts from the literature with actual regulatory outcomes, we can understand the correspondence and inconsistency between theory and practice.

3.4.1. Impact on innovation

There have been concerns about stifling innovation in AI regulations, but some studies contradict this assertion. The authors argued that well-designed rules can boost creativity by providing benchmarks that ensure safe and ethical AI development. In their view, the European AI Act may

enhance innovation because of the fixed conditions for AI developers of artificial intelligence (Tartaro et al., 2023). This contrasts with previous studies that saw impediments to regulation as a means of technological growth.

3.5. Advantages and disadvantages of AI regulation

AI regulation has both substantial benefits and notable challenges such as enhanced transparency and accountability, particularly in sensitive areas, such as healthcare, where algorithmic checks are required to limit prejudices and promote safety. For example, the EU AI Act, promotes trust among the general public by enforcing high-risk AI system compliance (Trilateral Research, 2023).

Challenges: It is commonplace for regulations to limit opportunities for innovation, especially with SMEs where the expenses and intricacy to comply is a disincentive to invest. Indian startups have articulated a focus on the economic burden of regulation which may inhibit progress in AI-led development (Economic Times, 2025).

3.6. SMEs and startups

AI regulation impacts innovation, particularly for SMEs and startups and has *Positive Effects*: Clear regulatory guidelines provide stability, encouraging investment and innovation in AI. The transparency requirements of the EU AI Act foster trust in AI applications and benefit developers. *Negative Effects*: Compliance costs disproportionately affect smaller enterprises, limiting their ability to innovate. For example, Indian startups face significant financial and operational burdens, underscoring the need for SME-friendly regulatory policies to balance accountability and growth potential.

3.6.1. Global regulatory standards

AI governance depends on the establishment of universal global standards. The authors claimed that international organizations should be involved in aligning AI regulations to enhance global cooperation and advancement. They suggest creating international safety protocols and regulatory systems to control the negative impacts of advanced artificial intelligence while simultaneously promoting global benefits (Ho et al., 2023). This corresponds with earlier research on globally recognized guidelines aimed at averting the fragmentation of regulation and ensuring uniformity across jurisdictions.

3.6.2. Anticipated challenges and criticisms

AI regulation is expected to face numerous anticipated challenges such as compliance, enforcement and unintended consequences. The article by Pagallo et al. (2022) points out the environmental challenges AI pose but criticizes the European AI Act for failing to address these concerns satisfactorily. To ensure sustainability, they opine that environmental impact assessments should be included in regulations related to artificial intelligence (Pagallo et al., 2022). This critique echoes earlier concerns identified in the literature about some regulatory frameworks being too narrow and the need for more holistic approaches.

3.6.3. Ethical and safety concerns

Thus, AI regulations should focus on ethical and safety concerns. Cooreman and Zhu (2022) discussed the difficulties surrounding AI definition and regulation in terms of ethics, proposing a multi-dimensional strategy that takes into consideration cultural settings as well as ecological issues. They endorse an AI governance model that is of relational character with the intention of providing all communities with appropriate mechanisms for compliance with their norms (Cooreman and Zhu, 2022). This perspective builds on previous discussions that pointed out the challenging issues regarding the regulation of ethical AI and the need for flexible contextually appropriate strategies.

3.6.4. Balancing regulation and innovation

A recurring theme is the modern-day balancing between regulation and innovation. According to Todorova et al. (2023), Ethical Innovation incorporates emerging technologies; however, their regulations need to be balanced. It is necessary to construct a proper system of regulations that assist small and medium-sized enterprises as well as start-ups that are more at risk than larger corporations (Todorova et al., 2023). This shift demonstrates a more sophisticated approach to deal with earlier suggestions for the framework regulating public interests and facilitating innovative activities.

3.6.5. Role of international collaboration

International cooperation is essential for the efficient governance of AI. This aspect is highlighted by Feijóo-Bandín et al. (2020), who make a case for technological diplomacy as a tool for achieving global consensus on collaborative governance and AI policy frameworks. They pointed out that this kind of collaborative engagement will promote better AI technology development and reduce regulation silos (Feijóo-Bandín et al., 2020). From their approach, there is some resemblance with previous literature that has argued for the benefits of collaborative initiatives on the global governance of AI.

3.6.6. Addressing regulatory gaps

It is crucial to adapt existing AI regulatory frameworks and fill these gaps. As detailed by Anderljung et al. (2023), there must be an effort to regulate frontier AI models that pose substantial risks to public safety. To help deal with these risks appropriately, they advocated for registration and compliance procedures, including standard-setting processes (Anderljung et al., 2023). This corresponds with the previously stated, more specific, and effective approaches that need to be taken with regulatory measures on the emerging AI risks.

This evaluation of the expected versus actual impacts of AI regulation indicates complex area where policy meets real-world functioning. One of the most important features of AI governance is the integration of open innovation approaches, which is characteristic of new governance systems. As a paradigm, Open Innovation is oriented towards building networks and pooling resources, which makes it indispensable for AI governance frameworks. Multi-stakeholder participation has already been incorporated in these frameworks, for example, the EU AI Act, to promote the responsible and transparent use of AI technologies. Regulations can incorporate more adaptive open innovation strategies that allow for greater flexibility and a more active form of governance. For instance, ***collaborative stakeholder engagement***: Open Innovation is conducive for the amalgamation of different perspectives from the government, universities, industries, and even civil societies as it allows participatory stakeholder engagement. This is akin to what (Chesbrough, 2003) advocated in his open innovation model where he encouraged the use of external knowledge and resources to achieve common objectives. ***The Ecosystem Approach to Governance Dynamic***: Stakeholder collaborative feedback through open ecosystems or diversified frameworks can further improve the risk classification and governance mechanisms for AI regulatory regimes. (Bresciani et al., 2018) illustrated how these frameworks can be developed using the collective intelligence of stakeholders.***Innovation-sensitive Regulation***: Open innovation-led regulatory approaches encourage both accountability and innovation through transparency. This enables enforcement obligations not to obstruct the advancement of technologies but instead foster innovation instead.

Integrating open innovation into AI governance serves not only as a tool for regulatory efficacy but also as a means of enhancing international collaboration in the development and management of AI technologies.

Although the European AI Act is expected to be instrumental in encouraging innovation and creating universal standards, it has inherent difficulties regarding observance, morals, and environmental consequences. To realize these twin goals, which facilitate inventiveness

while safeguarding everyone is interests, there must be continued fine-tuning and international joint efforts concerning AI regulations.

3.7. The EU AI Act

In terms of AI regulation, contemporary studies have sufficiently predicted the path, range, and characteristics of regulatory models such as that embodied in EU AI Act. Stepanyan (2021) gives a detailed examination of the EU's proposal on AI Act revealing its wide applicability and unified laws that concur with earlier scholarly debates about holistic policies towards AI systems (Stepanyan, 2021). Additionally, Hacker (2020) affirms the Act's emphasis on risk-oriented rules and transparency; these were key elements in past researches (Hacker, 2020).

With (Cordella and Gualdi, 2024) The approach taken by Italy in regulating generative AI illustrates problems with the technology-neutral approach. Such findings highlight the necessity for specially designed rules that mitigate the specific threats that generative AI brings. This is in line with the consideration of the broader need for regulatory flexibility and a context-sensitive approach as the technologies continue to change.

3.7.1. Risk-based approach and specific prohibitions

The risk-focused regulation of the EU AI Act matches well with earlier debates on AI system classification and expected limitations. EU AI Act based risk classification is shown in Fig. 2 below. The high-risk categorization within the Act is consistent with previous academic frameworks that classify AI systems by their possible risks and the requirement for strict regulatory scrutiny (Wagner et al., 2024). This strategy is also supported by Ebers (2021), who elaborate on the responsibilities and auditing protocols for highly hazardous artificial intelligence systems, echoing previous calls for tough risk mitigation strategies (Ebers, 2021).

3.7.2. Impact on innovation

The effect of AI regulation on innovation has been disputed. Although some believe that the EU AI Act would impede,but not foster, innovation, Tartaro et al. (2023) state that more stable and predictable conditions for AI development will be established, thus promoting innovation (Tartaro et al., 2023). This is in line with previous research, which has shown that clear regulatory frameworks can mitigate risks and encourage investment in AI technologies.

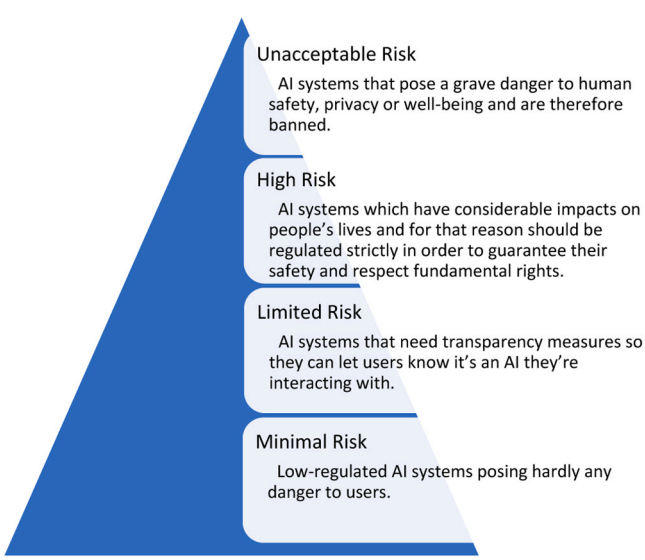

Fig. 2. Risk classification as per EU AI Act.

3.7.3. Global regulatory standards

The European Union Artificial Intelligence Act (EU AI Act) is becoming a global regulatory benchmark. Ho et al. (2023) argue that, doing so, will have an impact on international regulatory practices and encourage harmony while reducing fragmentation in regulation (Ho et al., 2023). This worldwide effect is consistent with previous studies that emphasised cooperative global moves to effectively govern AI technologies.

3.7.4. Anticipated challenges and criticisms

Although the EU AI Act has a comprehensive architecture, there are various expected problems and denunciation. Pagallo et al. (2022) assert that the Act does not take care of ecological matters properly and hence pleads for the infusion of sustainability principles in AI regulation (Pagallo et al., 2022). Gasser (2023) conversely looks at how continuous amendment of the Act will help address these dangers as well as the hybrid challenges raised by earlier proponents who called for regulatory frameworks that are adaptable and dynamic (Gasser, 2023).

3.7.5. Ethical and safety concerns

The EU AI Act focuses on ethical and safety issues, as in scholarly deliberations. In AI governance, Cooreman and Zhu (2022) suggested a relational style that factors cultural contexts and environmental impacts, such as those embedded in the EU AI Act for ethical reasons (Cooreman and Zhu, 2022). This ensures that socio-values can be implemented, thereby hence avoiding fundamental rights violations during the construction and deployment of AI systems.

3.7.6. Compliance and enforcement

There are many challenges to ensuring compliance with the EU's AI Act. For example, Walters et al. (2023) pointed out numerous compliance issues that affect organizations, which may depend on their scale and sector (Walters et al., 2023). These similarities only emphasize previous doubts concerning the effective application of AI principles by statutory provisions, and the need for helpful frameworks to ensure that the same guidelines are followed.

3.7.7. Further improvements needed

This is a positive development of the EU AI Act but there is still a need for further research. Sustainability impact assessments and clearer parameters for environmental sustainability should be included in the AI Act according to Hacker (2023). The scope of the Act should be extended to include lower-risk AI applications and ensure that all stakeholders influence AI development and monitor performance, as suggested by Salgado-Criado and Fernandez-Aller (2021).

The EU AI Act is well-suited to earlier academic forecasts concerning the manner, extent and characteristics of AI controls which include but are not limited to adapting risk-based approaches and having specific prohibitions. The law marks a turning point in terms of global regulatory standards as well as promoting innovative thinking, although there are also hurdles that need to be overcome, and criticisms will need to be dealt with in order for AI governance to become all-embracing and effective. In this way, it will be necessary for the whole framework to be revised continuously and an international partnership developed.

3.8. Findings & inferences

The swiftly developing and largely pervasive Artificial Intelligence (AI) is a subject of great debate in terms of the regulatory framework that can be used to mitigate its effects. Knowing the extent to which the existing literature anticipates AI regulation in terms of direction, extent and characteristics is important for aligning academic insights with current and future regulations. This systematic review intends to investigate how well theoretical discussions on AI regulation match with practical implementations in terms of concerns, objectives, governance mechanisms and the absence or presence thereof among the actual

regulations enacted or being formulated.

3.9. Research findings

3.9.1. Accuracy of literature predictions on AI regulation

Many aspects of AI regulation, including those contained in the EU AI Act, have been accurately predicted in the literature. On a broader scope, Stepanyan (2021) states that the Act is extensive and has harmonized rules as discussed earlier in academic discussions on the need for comprehensive regulations for AI systems (Stepanyan, 2021). Furthermore, according to Hacker (2023), this law emphasizes risk-based regulations and transparency which previous studies have also drawn attention to (Hacker, 2020). Article 1 of EU AI Act alligns with this and states that *"The AI Act will contain harmonized rules for placing on the market, putting into service, and using AI systems, ensuring compliance and accountability" (EU AI Act, Article 1)*.

3.9.2. Risk-based approach and specific prohibitions

The risk-based approach in the EU AI Act aligns well with earlier discussions of AI system classification and anticipated restrictions. Wagner et al. (2024) show how the high-risk classification in the Act follows previous academic frameworks that categorized AI systems according to their possible dangers (Wagner et al., 2024). This alignment is further supported by Ebers et al. (2021) who discussed obligations and conformity assessment processes for high-risk AI systems (Ebers, 2021). *This point of view is clearly reflected in Article 6 of EU AI Act "The Act categorizes AI systems into different risk tiers, with high-risk systems subject to stringent compliance requirements" (EU AI Act, Article 6)*.

3.9.3. Impact on innovation

The emergence of AI is believed to have an intricate relationship with innovations. Likewise, Tartaro et al. (2023) claim that the adoption of the EU AI Act may enhance stability and certainty within AI production thereby enhancing innovation (Tartaro et al., 2023). This view concurs with past literature, which proposes that well-defined regulatory structures can decrease unpredictability and promote further venturing into the field of artificial intelligence. *The perspective is clearly outlined in Recital 3 of the EU AI Act and states that "The Act aims to foster innovation by creating a predictable regulatory environment that supports the development and deployment of AI systems" (EU AI Act, Recital 3)*.

3.9.4. Global regulatory standards

The EU AI Act is about to be the global regulatory standards for all others. Ho et al. (2023) explained that it could define international regulation, thereby creating one voice and reducing fragmentation on the part of regulators across the world (Ho et al., 2023). The worldwide nature aligns with previous literature that has emphasized the need for unified international action to effectively govern AI technologies. *This perspective has been outlined in Article 2 of the EUAI Act and mentions that "The EU AI Act applies to a broad range of entities involved with AI systems within and outside the Union, aiming to create consistent standards across the EU, which could help promote global regulatory convergence and reduce fragmentation" (EU AI Act, Article 2)*

3.9.5. Anticipated challenges and criticisms

Despite the comprehensive framework, the EU AI Act is expected to encounter numerous challenges and criticism. For instance, Pagallo et al. (2022) identified environmental issues that are not well tackled by the Act, thereby urging the introduction of sustainability concepts into AI regulation (Pagallo et al., 2022). Gasser (2023) similarly discusses the refinement of this legislation in response to new dangers and hybrid difficulties (Gasser, 2023). Recital 5 of the *EU AI Act outlines this perspective and "The EU AI Act acknowledges the need for regulatory measures to address risks and protect public interests, while fostering innovation and ensuring the smooth functioning of the internal market" (EU AI Act, Recital 5)*.

3.10. Possible gaps

3.10.1. Ethical and safety concerns

Cooreman and Zhu (2022), on the contrary, claim that the AI Act of the EU is critical in solving ethical problems as well as safety measures, as seen in cultural contexts, and environmental implications are taken care of (Cooreman and Zhu, 2022). This calls for a relational method to AI governance that incorporates cultural contexts and environmental consequences while ensuring that AI systems are developed and deployed in a way that protects rights and community standards. This indicates a lack of cultural flexibility within the existing legal framework. *Article 13 of the EU AI Act supports these requirements and states that "The Act emphasizes transparency, fairness, and accountability to safeguard fundamental rights" (EU AI Act, Article 13)*.

3.10.2. Compliance and enforcement

The EU AI Act pose significant challenges to compliance issues. There are different forms of compliance barriers that organizations face as identified by Walters et al. (2023) which are dictated by likes of size and sector (Walters et al., 2023). This implies a practical gap in AI regulations implementation and necessitates supportive compliance frameworks to meet regulatory requirements. *Article 25 of the EU AI Act articulates that "The Act mandates strict compliance measures and provides for regular audits to ensure adherence" (EU AI Act, Article 25)*.

3.10.3. Further improvements needed

Adjustments should be made to the EU AI Act. Issues pertaining to sustainability should be addressed, and there should be a proper formulation of the environmental sustainability aspect of the AI Act (Hacker, 2020). Additionally, Salgado-Criado and Fernández-Aller (2021) recommend that lower risk AI use cases should also be included within the ambit of the Act so that all stakeholders will benefit from AI development, and be able to control its use (Salgado-Criado and Fernandez-Aller, 2021). *The EU AI Act stated "The Act aims to establish harmonized rules for high-risk AI systems while ensuring consistency with existing Union laws on data protection, consumer protection, and fundamental rights" in its Recital 9 (EU AI Act, Recital 9)*

Broadly, the EU AI Act has proven not to be absolutely wrong. Earlier academic estimations of the intended direction, scope, and features of AI regulation, such as listing bans and using a risk-based approach, seem to be accurate.The Act is likely to have an impact on global regulatory standards and spur innovation; however, there are various limitations and criticisms that should be resolved to achieve comprehensive and successful AI governance. Regulators must continually adjust their approaches to meet the evolving challenges, and thus, international cooperation is necessary so that the regulation around artificial intelligence can be enhanced. *In support of this view "The Act is intended to serve as a comprehensive regulatory framework that balances innovation with societal safeguards" (EU AI Act, Article 1)*.

4. Conclusion

4.1. Summary of key findings

This review article focuses on the AI regulation landscape with a special focus on theoretical deliberations and their relationship with practice. Although legal frameworks have become more aligned with theoretical models, immense discrepancies still exists. The paper is also critical of models based on AI risks that classify AI systems into categories based on the threats they can cause and then regulate those deemed as high risk. These findings are necessary to comprehend how AI technology advancement problems are addressed by contemporary means of AI system regulation.

4.2. Significance and implications

This review carries the implications of utmost importance for policy makers and legislators attempting to untangle the complex issues surrounding AI regulations. This review enhances the existing literature by demonstrating the extent to which current efforts by policymakers correspond with empirical projections and theoretical expectations. Therefore, this review provides guidance to policymakers who want to create or reform broad-based AI regulations that can be easily adjusted with technological changes. In other words, this study highlights the need for frameworks that reduce risks while also promoting more responsible development of AI technologies aimed at preserving social as well as moral standards.

4.3. Recommendations for adaptable and future-proof frameworks

In order to implement adaptability and future-proofing in the governing AI: *Adaptable Steps*: Devise automated systems that revise regulatory clauses set enabling amendments depending on the progress in technology. *Future-Proof Steps*: Requires regularly scheduled impact assessments with regards to AI, so that emerging difficulties and advancements are always considered. These steps could resolve issues present in the EU AI Act through greater resilience to the risks posed by the unknown, while simultaneously aiding holistic accountability for the future.

4.4. Limitations

Several limitations were identified by the review of existing studies, including addressing only high-risk AI applications that may not exhaustively represent the wider AI ecosystem, including low-risk scenarios. As a result, sufficient regulation could be lacking. The other thing that this review points out is that there are methodological challenges when it comes to aggregating diverse global regulatory perspectives that may completely overlook specific local or regional regulatory requirements and challenges.

4.5. Future research directions

This review creates a clear path for future research to explore the operational efficiency of existing AI regulatory frameworks through empirical studies. The focus beyond high-risk applications must be expanded to include regulatory impacts on lower-risk AI technologies that are now pervasive, as indicated by this statement. In addition, the dynamics of AI regulation in fostering innovation in small and medium-sized enterprises and startups should be examined by weighing regulation against innovation. In addition, lessons could be learned from exploring international regulatory cooperation to achieve more uniform global governance of AI technologies.

4.6. Final thoughts

In summary, the review argues that it is extremely important to develop AI policy frameworks that are responsive in nature and are based on empirical and theoretical evidence. The review ends with an observation that legislators should draft a code of ethics for AI that is not only comprehensive but also able to withstand changes in technology. Such policies in the management of AI should also ensure that the advantages of AI are reaped, while there is safe guarding of responsibility and openness in AI development to meet the expectations of society and the law. There is need to subject many legal and policy frameworks to periodic review and modification to respond to the new risks or global prospects relating to Artificial Intelligence.

4.7. A UN covenant and its wider perspective

Just as the Paris Agreement portrays the joint governance initiatives for broken climate assistance, it is vital to consider that a global AI supervision system would greatly profit from a covenant within the UN, which attempts to reconcile opposing AI approaches. In such a case, certain base level requirements must be established, independent of the level of growth of the country's technology. These interoperability measures would be beneficial for countries that may not have developed AI, such as emerging markets, developing economies, and economies under transition The same can be said for tackling issues such as accountability for unethical AI practices, data privacy, and cross jurisdiction AI operations where the UN would ensure minimum standards are met.

Funding

This research received no external funding.

Ethical statement

The authors declare that this manuscript is theirs, it has not been published or submitted elsewhere, and follows ethical standards of publishing. This study did not involve human or animal subjects, and thus no ethical approval was required. All co-authors have consented to the submission and accept responsibility for its content.

CRediT authorship contribution statement

Aishwarya T R: Methodology. **Bhat Pranav:** Writing – review & editing. **Pullela Phani Kumar:** Methodology. **Makkithaya Krishnamoorthi:** Methodology. **Hegde Nikhil Venkatraman:** Writing – review & editing, Formal analysis. **Ranga Bhatta Shreepathy:** Writing – review & editing. **Salu Staissy:** Writing – review & editing. **Shetty Dasharathraj K:** Data curation, Conceptualization. **Ranganathan Vijaya Arjunan:** Visualization, Validation, Project administration. **Cenitta D:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This manuscript was prepared using AI-driven tools to guarantee academic honesty by citing the proper papers, increasing understanding by increasing linguistic clarity, and providing comprehensive literature analysis. https://www.citethisforme.com were used to cite the work efficiently. Grammarly and Paperpal were used to examine the text for grammatical mistakes, typos, and punctuation errors. Additionally, they provided stylistic suggestions that aided in the professional drafting of this paper. The comprehension power of Quillbot was used to put across complicated ideas concisely while maintaining the original context and meaning. Scopus AI and Consensus.app, both intuitive and intelligent search tools, helped us to understand and enrich our insights with unprecedented speed and clarity. Scholarcy helped improve the pace of the process as it abstracted related academic articles and critical findings, thereby helping bring together existing research which let to identifying research gaps. We employed Turnitin software to account for plagiarism check.

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